

# Sampling Matters: An Empirical Study of Point Cloud Sampling Strategies for Brain MRI-Based Alzheimer’s Disease Classification

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**Abstract.** Alzheimer’s disease (AD) is a leading cause of dementia, with structural MRI widely used to assess atrophy in the hippocampus, ventricles, and cortex. 3D CNNs achieve strong classification performance but rely on dense volumetric processing that limits deployment in resource-constrained settings. Point cloud representations offer a lightweight alternative, yet the sampling step is often overlooked: most pipelines apply uniform or random sampling without considering voxel reliability. In brain MRI, boundary voxels contain mixed-tissue signals caused by partial volume effects, whereas interior voxels are more geometrically stable. This paper presents a systematic study of five sampling strategies for brain MRI-derived point clouds: uniform, random, confidence-aware, reliability-aware (proposed), and boundary-aware sampling. The proposed method applies a hard distance threshold to exclude boundary-adjacent voxels and retain stable interior points. Experiments on the ADNI dataset using PointNet and PointNet++ across point densities of 2,048, 4,096, and 8,192 show that reliability-aware sampling achieves the best overall performance, reaching 84.85% accuracy, 96.60% sensitivity, and 86.06% F1 with PointNet, and 86.36% accuracy with PointNet++. Boundary-aware sampling consistently performs worst, confirming that reliable interior regions provide more useful structural information than anatomical boundaries. The results further show that sampling quality has a greater effect on performance than point density across both architectures.

**Keywords:** Alzheimer’s disease · point cloud learning · reliability-aware sampling · confidence-aware sampling · neuroimaging · geometric deep learning

## 1 Introduction

Alzheimer’s disease is the most common cause of dementia, affecting an estimated 55 million people worldwide and projected to reach 139 million by 2050 [1]. Structural MRI captures progressive atrophy in the hippocampus, ventricles, and cortex, making it a primary tool for clinical assessment and a core neurodegeneration biomarker within the NIA-AA biological framework [8]. Deep learning has made strong inroads here: 3D CNNs applied directly to T1-weighted volumes report accuracies above 99% on controlled benchmarks [2], yet a thorough review of over 30 such studies found that more than half of those numbers were inflated by data leakage, with rigorously evaluated methods settling at balanced accuracy between 0.76 and 0.89 on independent test sets [18]. Recent architectures push further; HemiNet, for instance, exploits interhemispheric asymmetry to reach AUC 0.84 for MCI-to-AD conversion [11], but at upwards of 164 GFLOPs per inference the deployment gap in resource-constrained settings remains wide open.

The inefficiency has a structural cause. A standard brain volume contains a large fraction of voxels with little or no diagnostic signal, yet volumetric models process every voxel identically. Representing the brain as a sparse set of points sampled from its diagnostically informative regions preserves the geometry that matters for classification while discarding redundant spatial information, at a fraction of the cost [14,15]. Our prior work introduced Anatomical Priority Sampling (APS) and the ADNI-2DPC dataset, showing that anatomically guided point cloud generation achieves competitive AD classification with substantially lower inference latency [6], and in doing so flagged the sampling strategy itself as a variable worth examining more carefully.

Standard pipelines sample points uniformly or at random without consideration of the structural properties of those regions, an assumption that is questionable in neuroimaging: voxels near tissue boundaries carry mixed signals from partial volume effects and scanner noise, making them unreliable as geometric descriptors [3]. When segmentation precedes point cloud generation, boundary-proximal points may not reliably belong to the region they were assigned to, introducing a second layer of geometric ambiguity [4,12]. Interior points, drawn from within clearly defined anatomical regions, tend to be more geometrically consistent and carry more discriminative spatial information. The impact of this interior-versus-boundary distinction on point cloud quality and downstream classification performance has not been studied systematically.

Despite the growing body of work on point cloud architectures for medical imaging, almost all of it focuses on improving model design while treating the upstream sampling step as fixed [21]. No prior work has conducted a controlled comparison of sampling strategies designed to examine the relationship between anatomical reliability and downstream classification performance in brain MRI-derived point clouds. This paper addresses that gap through the following contributions:

1. A systematic empirical comparison of five sampling strategies (uniform, random, confidence-aware, reliability-aware threshold (proposed), and boundary-aware) for brain MRI-derived point clouds.
2. A reliability-aware threshold strategy that excludes boundary-proximal voxels and retains only stable interior points, paired with a contrastive boundary-aware baseline to isolate the role of interior reliability.
3. An ROI-aware pipeline converting T1-weighted MRI into a fixed-size, anatomically labelled 2D point cloud over four AD-relevant regions (hippocampus, ventricles, cortex, interior) via heuristic priors, avoiding atlas registration.
4. Evaluation across three densities ( $N \in \{2,048, 4,096, 8,192\}$ ) and two backbones (PointNet and PointNet++), showing that sampling reliability outweighs point density and generalises across architectures.

## 2 Related Work

We review four areas directly relevant to this work: MRI-based AD classification, medical point cloud learning, point cloud sampling strategies, and boundary reliability in medical imaging.

### 2.1 MRI-Based Alzheimer’s Disease Classification

The use of deep learning on structural MRI for AD classification has a decade-long track record [10]. End-to-end CNNs quickly replaced handcrafted morphometric features: Basaia et al. showed that a 3D all-convolutional network trained on ADNI T1-weighted MRI, without manual feature engineering, achieves 99.2% accuracy for AD versus healthy control discrimination across imaging protocols and sites [2]. Wen et al. subsequently reviewed over 30 such studies and found that the majority likely overstated performance through data leakage; when evaluated rigorously, balanced accuracy on independent test sets ranged from 0.76 to 0.89 [18]. More recently, HemiNet exploits interhemispheric structural asymmetry through an attention-guided 3D CNN, reaching AUC 84.01% on ADNI and 81.37% on an external cohort [11], yet at 164.53 GFLOPs it exemplifies the persistent computational demands of volumetric approaches. All these methods share a common constraint: they process every voxel regardless of clinical relevance, motivating exploration of more compact representations such as point clouds that encode only anatomically meaningful structure at substantially lower cost.

### 2.2 Medical Point Cloud Learning

PointNet [14] was the first architecture to process raw unordered point clouds end-to-end, using shared MLP layers and global max pooling to achieve permutation invariance for classification and segmentation. PointNet++ [15] extended this with hierarchical neighbourhood aggregation, capturing local geometry at multiple scales and improving robustness to density variation. Both remain the

standard baselines in medical point cloud work. Medical applications have since expanded: Yang and Gao introduced PointCHD for congenital heart disease classification [20]; Mekhzoum et al. showed PointNet-based descriptors transfer to 4D-CT joint motion registration at 2.26 mm mean error [13]; GRAB-Net demonstrated that modelling boundary ambiguity through graph-based reasoning yields consistent segmentation gains on tooth and aneurysm datasets [12]; and Zhang et al. showed complementary gains from multi-scale feature fusion for dental point cloud completion [21]. For brain MRI, our prior work produced the ADNI-2DPC dataset and showed that anatomically guided sampling achieves competitive AD classification at substantially lower latency than volumetric baselines [6]. A limitation runs through all of this work: the point cloud generation step is treated as fixed preprocessing, with little attention paid to how the sampling strategy affects the quality of the resulting representation.

### 2.3 Point Cloud Sampling Strategies

Sampling determines what information enters the network. Uniform sampling treats every spatial location as equally informative [14]; random sampling introduces stochasticity and scales well to large outdoor scenes paired with local feature aggregation [5]; and farthest point sampling (FPS), used by PointNet++ to build local neighbourhoods, maximises spatial coverage iteratively [15], though coverage and reliability are not the same thing. More sophisticated strategies include PointASNL, which perturbs FPS-selected coordinates based on local context to improve outlier robustness [19]; KPConv, which enforces consistent density through grid subsampling [16]; and DGCNN, which dynamically recomputes  $k$ -nearest neighbour graphs in feature space to capture semantically meaningful local groupings, achieving 92.9% accuracy on ModelNet40 [17]. These are genuine advances for general 3D scenes, but none was designed with anatomical reliability in mind, and none has been evaluated in neuroimaging, where partial volume effects and segmentation uncertainty create a qualitatively different noise source at region boundaries, which is the gap this work addresses directly.

### 2.4 Boundary and Reliability in Medical Imaging

Boundary-related unreliability in medical imaging is well documented, though it has rarely been carried through to the point cloud generation stage. Partial volume effects cause voxels near tissue interfaces to mix signal from multiple tissue types, producing geometrically ambiguous measurements less stable than interior voxels [3]. Billot et al. showed this degrades standard segmentation severely across diverse MRI protocols [3], and when segmentation precedes point cloud generation, that instability carries directly into the representation. Ghazvinian Zanjani et al. found that concentrating sampling in reliable interior regions lifted IoU from 0.88 to 0.91 on dental point cloud segmentation [4], an early signal that where you sample matters, not just how many points you take. GRAB-Net reinforced this: boundary-proximal points introduce feature ambiguity that measurably degrades segmentation quality, while separating interior from boundary

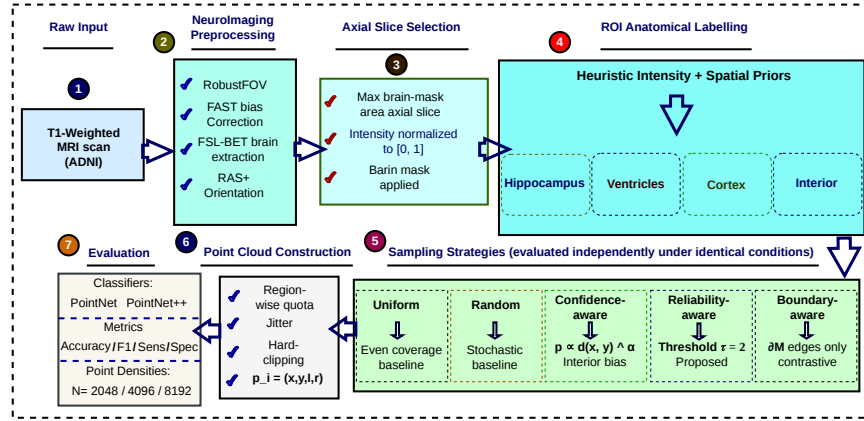
regions consistently improves it [12]. Boundary points are not merely noisier; they are structurally less reliable as descriptors, and excluding them yields a cleaner, more discriminative representation.

In summary, boundary points in medical point clouds are structurally less reliable than interior points [3,4,12], yet no prior work has used this contrast to guide sampling for brain MRI-based AD classification. The present study addresses that gap directly..

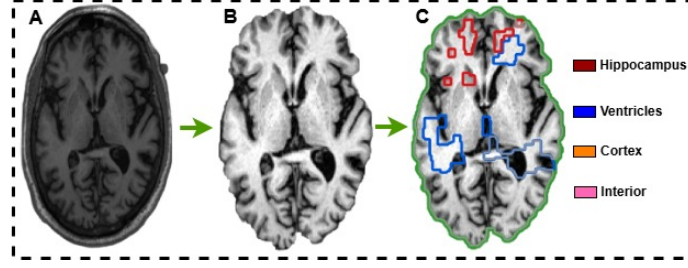
### 3 Methodology

#### 3.1 Framework Overview

The proposed framework, illustrated in Fig. 1, translates a T1-weighted MRI scan into a fixed-size, anatomically labelled 2D point cloud and then feeds it to a geometric deep learning classifier. The framework consists of four high-level phases comprising seven stages that executed in sequence: standard neuroimaging preprocessing, heuristic-ROI region generation, sampling-strategy-specific point cloud construction, and downstream classification. Each sampling strategy operates on the same preprocessed image and populates the same region-wise quota; only the rule by which candidate pixels are selected within each region changes. This design keeps every other variable constant and makes the downstream accuracy difference attributable to sampling alone.



**Fig. 1.** End-to-end pipeline of the proposed reliability-aware point cloud generation framework for Alzheimer’s disease classification. A T1-weighted MRI scan is pre-processed (Stages 1–3), labelled into four AD-relevant regions (Stage 4), converted to a point cloud under five sampling strategies evaluated independently (Stage 5), and assembled into the final representation  $p_i = (x, y, I, r)$  (Stage 6) for classification with PointNet and PointNet++ across point densities  $N \in \{2,048, 4,096, 8,192\}$  (Stage 7).



**Fig. 2.** Stage-wise visualisation of the data preparation pipeline on a representative subject. **(A)** Raw T1-weighted MRI input. **(B)** Brain-only volume after preprocessing (Stage 2, Fig. 1). **(C)** Heuristic ROI labelling of four AD-relevant regions used for downstream sampling (Stage 4, Fig. 1).

### 3.2 MRI Preprocessing

Each scan undergoes a standard preprocessing chain before any sampling takes place (Stage 2, Fig. 1). **RobustFOV** crops the field of view to remove non-brain tissue, **FSL-FAST** corrects low-frequency bias-field inhomogeneities, and **FSL-BET** extracts the brain volume. The resulting mask is refined by morphological opening, closing, and hole-filling, then reoriented to  $RAS^+$ .

A single representative 2D axial slice is then selected from the extracted volume (Stage 3, Fig. 1). The slice with the maximum brain-mask area is chosen, consistently capturing the hippocampus, lateral ventricles, and cortical mantle in a single plane. Pixel intensities are normalised to  $[0, 1]$  using within-mask percentiles  $p_1$  and  $p_{99}$ :

$$I'(x, y) = \frac{I(x, y) - p_1}{p_{99} - p_1 + \epsilon}, \quad \epsilon = 10^{-8}, \quad (1)$$

and pixels outside the brain mask are set to zero.

### 3.3 ROI Region Generation

Rather than relying on atlas-based segmentation, which introduces external dependencies and limits reproducibility across scanners, this work derives approximate anatomical masks directly from intensity and spatial priors on the normalised 2D slice (Stage 4, Fig. 1; Fig. 2). Four non-overlapping regions are defined: hippocampus, ventricles, cortex/brain surface, and interior white matter.

**Ventricles.** Candidate voxels lie inside a central ellipse of radii  $0.35h$  and  $0.35w$  with intensities below the 20th or above the 80th within-brain percentile. The mask is refined by binary opening ( $5 \times 5$ ), closing ( $7 \times 7$ ), and hole-filling, retaining the three largest connected components.

**Hippocampus.** Candidate pixels lie within bilateral medial-temporal bands ( $|x - c_x| > 0.12w$ ,  $c_y - 0.2h < y < c_y + 0.25h$ ), fall outside the ventricular mask, and have intensities in the 35th–85th percentile range. After morphological refinement, the six components closest to the median size are retained to capture bilateral hippocampal heads and tails.

**Cortex / brain surface.** The cortical shell is obtained by subtracting a morphologically eroded brain mask from the full mask, excluding voxels already assigned to the hippocampus or ventricles.

**Interior.** All remaining brain-mask voxels are assigned to the interior class, covering white matter and deep grey-matter structures not captured by the three heuristics above.

### 3.4 Point Cloud Representation

Each scan is represented as a set  $\mathcal{P} = \{p_i\}_{i=1}^N$  of  $N$  unordered points drawn from the 2D slice. Each point is a four-dimensional feature vector,

$$p_i = (x_i, y_i, I_i, r_i), \quad (2)$$

where  $(x_i, y_i)$  are spatial coordinates in millimetres,  $I_i \in [0, 1]$  is the normalised MRI intensity, and  $r_i \in \{0, 1, 2, 3\}$  is the anatomical region label (interior, hippocampus, ventricle, cortex). Pixel indices are scaled by the NIfTI voxel spacings, making the representation invariant to scanner resolution. The total point budget  $N \in \{2,048, 4,096, 8,192\}$  is split across regions using fixed quotas: 35% hippocampus, 25% ventricles, 20% cortex, and 20% interior. At  $N = 8192$  this yields 2,867 hippocampal, 2,048 ventricular, 1,638 cortical, and 1,639 interior points per scan. The hippocampus is over-weighted relative to its voxel fraction, reflecting its role as the most established neuroimaging biomarker for Alzheimer’s disease [9].

### 3.5 Sampling Strategies

Five sampling strategies are evaluated independently under identical conditions (Stage 5, Fig. 1). The first two are baselines that distribute points without regard to anatomical reliability; the next two are interior-biased strategies, with the reliability-aware threshold as the proposed method; the fifth is a contrastive boundary-focused baseline.

**Uniform Sampling.** Points are drawn with uniform probability from within each region mask, treating every voxel as equally informative regardless of its proximity to segmentation boundaries.

**Random Sampling.** Points are drawn uniformly at random from the entire brain mask, independently of region membership, with labels  $r_i$  assigned post-hoc. This replicates the common default in point cloud pipelines and serves as the stochastic baseline.

**Confidence-Aware Interior Sampling.** Each candidate is weighted by its squared Euclidean distance from the nearest region boundary, on the premise that distance from the boundary correlates with anatomical certainty. For region mask  $\mathcal{M}$ , the selection probability is:

$$\Pr[\text{select } (x, y)] \propto [\text{EDT}(\mathcal{M})(x, y)]^\alpha, \quad \alpha = 2. \quad (3)$$

Squaring the distance suppresses boundary voxels and concentrates sampling mass in the geometrically stable interior.

**Reliability-Aware Threshold Sampling (Proposed).** The proposed strategy replaces the soft weighting above with a hard geometric criterion: a voxel is eligible only if its distance from the nearest boundary meets or exceeds a threshold  $\tau$ ,

$$\mathcal{M}_\tau = \{(x, y) \in \mathcal{M} : \text{EDT}(\mathcal{M})(x, y) \geq \tau\}, \quad (4)$$

with  $\tau = 2\text{px}$  in all experiments. Sampling is then uniform over  $\mathcal{M}_\tau$ ; if  $\mathcal{M}_\tau$  is empty, the full mask is used as a fallback. The hard exclusion zone produces a cleaner pool of anatomically stable voxels and, as Section 5 shows, delivers the best classification performance across all point budgets.

**Boundary-Aware Sampling.** This strategy inverts the criterion: candidates are restricted to the structural boundary alone,

$$\partial\mathcal{M} = \mathcal{M} \setminus \text{erode}(\mathcal{M}, 3 \times 3), \quad (5)$$

focusing exclusively on atrophy-associated hippocampal contours and ventricular enlargement edges. It serves as a contrastive baseline that quantifies the information value of boundaries versus reliable interiors.

**Region-Wise Quota Allocation and Final Construction.** All five strategies share the same closing pipeline (Stage 6, Fig. 1). Per-region point sets are concatenated, small jitter ( $\pm 0.15\text{ mm}$  for surface regions,  $\pm 0.20\text{ mm}$  for interior) is applied when replacement sampling is needed, and hard clipping removes any point outside the brain mask. The resulting fixed-size cloud of  $N$  points is passed to the classifier.

## 4 Experimental Setup

### 4.1 Dataset and Implementation Details

All experiments use T1-weighted MRI scans from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) [7]. The working set comprises 700 subjects (350 AD, 350 CN) with a subject-level 80:20 stratified split. The same split is used across all five sampling strategies and three point budgets, so performance differences reflect the sampling design alone. Each pipeline is applied independently per scan, producing five parallel point cloud datasets from the same MRI volumes.

All models are implemented in PyTorch and trained on a single NVIDIA RTX 3060 GPU (12 GB VRAM). Training uses the Adam optimiser ( $\text{lr} = 10^{-3}$ , weight decay =  $10^{-4}$ , batch size 16) for up to 150 epochs with early stopping after 20 epochs of no validation improvement; the best checkpoint is retained. Augmentation includes coordinate jittering ( $\sigma = 0.01$ ), uniform scaling ( $[0.95, 1.05]$ ), and intensity jittering ( $\sigma = 0.02$ ); rotation is excluded because MRI scans are acquired in a canonical orientation and in-plane rotation would corrupt the anatomical region labels. A fixed random seed of 42 is used throughout. Inference latency is measured as wall-clock milliseconds per sample over

30 batches after a 10-batch warm-up, and peak GPU memory is tracked via `torch.cuda.max_memory_allocated`.

## 4.2 Architecture, Evaluation Metrics, and Experimental Protocol

PointNet [14] serves as the primary classifier, processing the four-dimensional input  $p_i = (x, y, I, r)$  through shared MLP layers and global max-pooling to produce a permutation-invariant descriptor for binary AD/CN prediction. PointNet++ [15] is additionally evaluated to test whether the sampling effect generalises to a hierarchical architecture. Performance is reported using accuracy, f1-score, sensitivity (AD recall), and specificity (CN recall), with inference latency and peak GPU memory included to characterise computational cost across point densities.

Five sampling strategies across three point densities ( $N \in \{2048, 4096, 8192\}$ ) yield 15 experimental configurations, each trained independently with fresh random initialisation. For densities below 8192, the stored 8192-point cloud is sub-sampled on-the-fly without replacement, so the same spatial point set is used across densities for a given strategy. The best validation checkpoint is used throughout.

# 5 Results and Analysis

## 5.1 Overall Classification Performance

Table 1 reports classification performance of five sampling strategies evaluated with PointNet architecture at a fixed point density of 8,192 points, the highest density tested. The proposed reliability-aware threshold strategy achieves the strongest performance: 84.85% accuracy, 96.60% sensitivity, and 86.06% F1. The high sensitivity is clinically meaningful, since missing an AD case carries greater cost than a false alarm in screening. Confidence-aware sampling follows at 83.83% accuracy and 91.50% sensitivity, showing that interior-biased strategies outperform boundary-focused and uniform baselines. Random sampling reaches the highest specificity (85.29%), reflecting better identification of cognitively normal subjects but at the cost of sensitivity and overall accuracy. Boundary-aware sampling records the lowest accuracy and F1, indicating that restricting candidates to structural contours degrades discriminative power despite producing a geometrically distinct representation.

## 5.2 Effect of Sampling Strategy

Since the model, data split, and training protocol remain fixed, the performance differences in Table 1 are driven solely by the sampling strategy. Interior-focused approaches perform best: reliability-aware and confidence-aware sampling improve accuracy and F1 by 3–5% over uniform and random baselines, and by

**Table 1.** Comparative classification performance of five point cloud sampling strategies at  $N = 8192$  using PointNet. Best result in each column in **bold**.

| Strategy                 | Acc (%)      | Sens (%)     | Spec (%)     | F1 (%)       |
|--------------------------|--------------|--------------|--------------|--------------|
| Uniform                  | 80.30        | 81.25        | 79.41        | 80.00        |
| Random                   | 81.82        | 78.12        | <b>85.29</b> | 83.64        |
| Confidence-Aware         | 83.83        | 91.50        | 82.41        | 83.58        |
| <b>Reliability-Aware</b> | <b>84.85</b> | <b>96.60</b> | 84.50        | <b>86.06</b> |
| Boundary-Aware           | 78.79        | 75.00        | 82.35        | 77.42        |

6–9% over boundary-aware sampling. The consistent F1 ranking further confirms this trend across metrics. The poor performance of boundary-aware sampling is particularly notable. Although it explicitly targets hippocampal contours and ventricular edges, boundary voxels often contain mixed-tissue signals caused by partial volume effects. This ambiguity introduces noise directly into the point cloud, weakening the resulting representation. Among the interior-focused methods, reliability-aware sampling ( $\tau = 2\text{px}$ ) slightly outperforms confidence-aware weighting, achieving 1% higher accuracy and 2.5% higher F1. Excluding boundary-proximal voxels entirely proves more effective than downweighting them, because even low-probability boundary samples accumulate at  $N = 8192$  to degrade the representation.

### 5.3 Ablation Study

With the model, training protocol, and data split fixed, Table 1 isolates the effect of each sampling design choice.

**Hard threshold vs. soft weighting.** Reliability-aware and confidence-aware sampling differ in one respect: a soft weighting ( $p \propto d^\alpha$ ) versus a hard exclusion zone ( $d \geq \tau$ ). This single change improves accuracy from 83.83% to 84.85%, F1 from 83.58% to 86.06%, and sensitivity from 91.50% to 96.60%, confirming that excluding boundary voxels entirely outperforms downweighting them.

**Interior vs. boundary selection.** Reversing the rule to sample only from boundaries gives the opposite result. Boundary-aware sampling drops to 78.79% accuracy and 77.42% F1, a degradation of 6.06% and 8.64% relative to reliability-aware, confirming that the gain comes from interior reliability.

**Takeaway.** Both ablations identify interior reliability as the source of improvement, and the hard distance threshold as the most effective way to enforce it.

### 5.4 Effect of Point Density

Each sampling strategy is evaluated at three point budgets  $N \in \{2,048, 4,096, 8,192\}$ . Table 2 reports accuracy, sensitivity, specificity, F1, inference latency, and peak GPU memory.

**Interior-focused strategies are more point-efficient.** Reliability-aware and confidence-aware sampling achieve 83.33% and 81.88% accuracy even at 2,048

**Table 2.** Effect of point density on classification performance and computational cost across five sampling strategies (PointNet). Best performance per strategy in **bold**.

| Strategy                 | N    | Acc          | Sens         | Spec         | F1           | Lat  | GPU   |
|--------------------------|------|--------------|--------------|--------------|--------------|------|-------|
| Uniform                  | 2048 | 78.79        | 87.50        | 70.59        | 80.00        | 6.72 | 152.6 |
|                          | 4096 | 78.79        | <b>92.88</b> | 71.76        | <b>81.58</b> | 7.01 | 282.7 |
|                          | 8192 | <b>80.30</b> | 81.25        | <b>79.41</b> | 80.00        | 7.54 | 543.0 |
| Random                   | 2048 | 80.83        | 87.50        | 79.41        | 80.58        | 6.75 | 152.6 |
|                          | 4096 | 80.63        | <b>87.50</b> | 79.47        | 82.35        | 7.05 | 282.7 |
|                          | 8192 | <b>81.82</b> | 78.12        | <b>85.29</b> | <b>83.64</b> | 7.48 | 543.0 |
| Confidence-Aware         | 2048 | 81.88        | 88.88        | 67.65        | 83.29        | 6.89 | 152.6 |
|                          | 4096 | 82.85        | 90.38        | 82.35        | 83.08        | 7.10 | 282.7 |
|                          | 8192 | <b>83.83</b> | <b>91.50</b> | <b>82.41</b> | <b>83.58</b> | 7.52 | 543.0 |
| <b>Reliability-Aware</b> | 2048 | 83.33        | 90.90        | 70.60        | 84.95        | 8.02 | 151.0 |
|                          | 4096 | 84.85        | 93.80        | 82.50        | 85.75        | 7.86 | 279.6 |
|                          | 8192 | <b>84.85</b> | <b>96.60</b> | <b>84.50</b> | <b>86.06</b> | 7.77 | 536.7 |
| Boundary-Aware           | 2048 | 74.24        | 46.88        | 81.18        | 60.01        | 6.71 | 152.6 |
|                          | 4096 | 75.76        | 68.75        | 82.35        | 73.33        | 7.02 | 282.7 |
|                          | 8192 | <b>78.79</b> | <b>75.00</b> | <b>82.35</b> | <b>77.42</b> | 7.45 | 543.0 |

points, while uniform sampling requires 8,192 points to reach 80.30%. This indicates that interior-biased sampling extracts more informative features per point. **Cost scales linearly and stays low.** Moving from  $N = 2,048$  to  $N = 8,192$  raises latency by about 12% and GPU memory by roughly  $3.5\times$ , consistent with PointNet’s linear complexity. Even the largest configuration remains practical, requiring under 8 seconds per validation pass and about 543 MB of GPU memory. **More boundary points do not resolve ambiguity.** Although boundary-aware sampling benefits most from higher density (74.24% to 78.79% accuracy), it consistently underperforms all interior-focused methods. Increasing the number of boundary samples reduces variance but cannot remove the mixed-tissue ambiguity inherent to boundary voxels.

### 5.5 Generalisation to Model Architecture

The same five strategies are evaluated with PointNet++ to test whether the sampling effect generalises across backbones (Table 3).

**Reliability-aware sampling remains the strongest strategy.** The proposed method achieves the highest accuracy on both backbones, reaching 84.85% with PointNet and 86.36% with PointNet++ at  $N = 8192$ . It also attains the best sensitivity and F1 on PointNet, and the highest specificity on PointNet++, indicating a more balanced classifier overall.

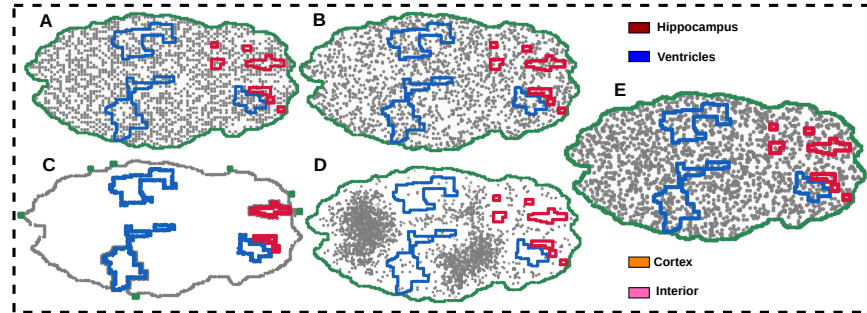
**Other strategies show metric trade-offs.** Confidence-aware, boundary-aware, and random sampling each perform well on isolated metrics, but none match the overall consistency of reliability-aware sampling across architectures and evaluation measures.

**Table 3.** Performance of five sampling strategies on PointNet and PointNet++ across all point budgets. Highest Acc, Sens, Spec, and F1 per backbone in **bold**.

| Strategy    | N    | PointNet     |              |              |              |      |       | PointNet++   |              |              |              |      |        |
|-------------|------|--------------|--------------|--------------|--------------|------|-------|--------------|--------------|--------------|--------------|------|--------|
|             |      | Acc          | Sens         | Spec         | F1           | Lat  | GPU   | Acc          | Sens         | Spec         | F1           | Lat  | GPU    |
| Uniform     | 2048 | 78.79        | 87.50        | 70.59        | 80.00        | 6.72 | 152.6 | 80.82        | 84.40        | 79.40        | 81.83        | 6.98 | 553.1  |
|             | 4096 | 78.79        | 92.88        | 71.76        | 81.58        | 7.01 | 282.7 | 82.85        | 84.40        | 80.30        | 84.39        | 7.20 | 1066.5 |
|             | 8192 | 80.30        | 81.25        | 79.41        | 80.00        | 7.54 | 543.0 | 82.85        | <b>93.80</b> | 86.50        | 85.75        | 7.78 | 3373.5 |
| Random      | 2048 | 80.83        | 87.50        | 79.41        | 80.58        | 6.75 | 152.6 | 81.33        | 83.80        | 73.50        | 84.52        | 6.84 | 553.1  |
|             | 4096 | 80.63        | 87.50        | 79.47        | 82.35        | 7.05 | 282.7 | 81.85        | 84.40        | 85.30        | 84.39        | 6.90 | 1066.5 |
|             | 8192 | 81.82        | 78.12        | <b>85.29</b> | 83.64        | 7.48 | 543.0 | 83.88        | 87.50        | 88.20        | 85.48        | 7.39 | 3373.5 |
| Confidence  | 2048 | 81.88        | 88.88        | 67.65        | 83.29        | 6.89 | 152.6 | 80.30        | 87.50        | 73.50        | 81.15        | 6.88 | 553.1  |
|             | 4096 | 82.85        | 90.38        | 82.35        | 83.08        | 7.10 | 282.7 | 83.33        | 88.10        | 86.50        | <b>86.92</b> | 7.12 | 1066.5 |
|             | 8192 | 83.83        | 91.50        | 82.41        | 83.58        | 7.52 | 543.0 | 83.33        | 89.00        | 89.40        | 86.19        | 7.42 | 3373.5 |
| Reliability | 2048 | 83.33        | 90.90        | 70.60        | 84.95        | 8.02 | 151.0 | 83.33        | 84.00        | 85.30        | 84.23        | 7.99 | 1226.7 |
|             | 4096 | <b>84.85</b> | 93.80        | 82.50        | 85.75        | 7.86 | 279.6 | 84.85        | 84.90        | 94.10        | 85.71        | 7.93 | 1739.0 |
|             | 8192 | <b>84.85</b> | <b>96.60</b> | 84.50        | <b>86.06</b> | 7.77 | 536.7 | <b>86.36</b> | 93.40        | <b>97.10</b> | 86.39        | 8.10 | 4046.0 |
| Boundary    | 2048 | 74.24        | 46.88        | 81.18        | 60.01        | 6.71 | 152.6 | 81.82        | 84.40        | 82.40        | 83.11        | 6.63 | 553.1  |
|             | 4096 | 75.76        | 68.75        | 82.35        | 73.33        | 7.02 | 282.7 | 81.82        | 84.80        | 79.40        | 83.98        | 6.63 | 1066.5 |
|             | 8192 | 78.79        | 75.00        | 82.35        | 77.42        | 7.45 | 543.0 | 82.82        | 88.10        | 85.30        | 85.63        | 7.18 | 3373.5 |

**Computational cost remains practical.** Although PointNet++ increases GPU memory by 2–7 $\times$  over PointNet, latency stays below 8.1 seconds in all settings. The largest reliability-aware configuration requires about 4.0 GB of GPU memory, remaining feasible on a single consumer-grade GPU.

## 5.6 Qualitative Analysis



**Fig. 3.** Point cloud distributions for a representative AD subject at  $N = 8,192$ . Colours denote anatomical regions as labelled. (A) Uniform sampling. (B) Random sampling. (C) Boundary-aware sampling. (D) Confidence-aware sampling. (E) Reliability-aware sampling (proposed).

Fig. 3 shows the point cloud distributions for a representative AD subject at  $N = 8,192$ . Panels A and B are visually near-identical, consistent with their similar quantitative performance. Both distribute points evenly without distinguishing boundary from interior voxels. Panel C (boundary-aware) collapses to a thin

contour, leaving the interior empty and confirming why this strategy produces the weakest classifier. Panel D (confidence-aware) shows dense interior coverage but with residual clustering near region edges, where the soft  $d^\alpha$  weighting still admits boundary-proximal samples. Panel E (proposed reliability-aware) is the most uniform of the five. The hard threshold ( $\tau = 2\text{px}$ ) enforces a clean exclusion zone around every boundary, retaining only stable interior voxels. This directly reflects the strongest result in our experiments: 84.85% accuracy and 96.60% sensitivity on PointNet, and 86.36% accuracy on PointNet++.

## 5.7 Discussion

The main finding of this study is that point cloud quality matters more than point count. At  $N = 8192$ , reliability-aware sampling improves PointNet accuracy and F1 by 4.55% and 6.06% over uniform sampling, while boundary-aware sampling reduces both metrics by 6–9%. The qualitative results in Fig. 3 support this trend: reliability-aware sampling produces cleaner interior coverage, whereas confidence-aware sampling still retains boundary-adjacent points.

Reliability-aware sampling also achieves the highest sensitivity (96.60%) while maintaining specificity above 84%, giving a more balanced operating profile for AD screening. The method remains computationally practical, with latency below 8 seconds and peak GPU memory around 4 GB on PointNet++. It is also point-efficient, as reliability-aware sampling at  $N = 2048$  already matches the performance of uniform sampling at  $N = 8192$ .

The same sampling trend is preserved across PointNet and PointNet++, with PointNet++ improving reliability-aware accuracy by 1.5%. Both backbones were originally developed for natural-object recognition, where discriminative features are mainly surface-based, whereas brain MRI point clouds are highly stereotyped and differ primarily through subtle volumetric atrophy. The 86.36% accuracy achieved with a general-purpose backbone therefore reflects the contribution of the sampling strategy itself, while neuroimaging-specific architectures remain an important direction for future work.

Overall, the study shows that a simple hard-distance threshold can produce cleaner point clouds and improve AD classification with minimal computational cost. Future work should evaluate the method on larger cohorts, MCI progression tasks, and neuroimaging-specific point cloud architectures.

## 6 Conclusion

Five sampling strategies were evaluated under identical conditions on the ADNI dataset to isolate the effect of sampling on brain MRI-derived point cloud classification for Alzheimer’s disease. Reliability-aware threshold sampling achieved the best results across all experiments: 84.85% accuracy, 96.60% sensitivity, and 86.06% F1 on PointNet at  $N = 8192$ , and 86.36% accuracy with 97.10% specificity on PointNet++. The boundary-aware contrastive experiment confirmed that the gain comes from interior reliability, as reversing the criterion dropped accuracy by 6.06% and F1 by 8.64%. Reliability-aware sampling at  $N = 2048$

matched uniform sampling at  $N = 8192$ , showing that sampling strategy influences performance more than point count. The same pattern held across both backbones, confirming that the benefit arises from the input representation rather than model capacity. A simple hard-distance threshold produces cleaner point clouds and stronger AD classification at no additional inference cost, and its simplicity makes it directly applicable to other neuroimaging pipelines and a foundation for neuroanatomy-specific architectures.

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